

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Electrical and Computer Engineering

LAB REPORT-02

Course Code : ECE 2112

Course Title : Digital Techniques Sessional

Experiment Name : Boolean Function Simplification Using Boolean Algebra and Verification Using Logic Gates

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Task 1: Simplification and Implementation of $F(A, B, C) = A'BC + AB'C + ABC' + ABC$

Objectives

- To simplify the given Boolean expression using Boolean algebra.
- To implement the unsimplified and simplified expressions as logic circuits.
- To verify that both circuits produce identical outputs for all input combinations.

Apparatus

Digital logic circuit simulation software, AND/OR/NOT gate blocks, input switches, output indicator (LED).

Given Expression

$$F(A, B, C) = A'BC + AB'C + ABC' + ABC$$

Simplification

$$\begin{aligned} F(A, B, C) &= A'BC + AB'C + ABC' + ABC \\ &= A'BC + ABC + AB'C + ABC + ABC' + ABC \\ &= BC(A' + A) + AC(B' + B) + AB(C' + C) \\ &= BC + AC + AB \\ &= AB + AC + BC \end{aligned}$$

Truth Table Verification

A	B	C	$F = AB + AC + BC$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Circuit Design

The unsimplified expression $A'BC + AB'C + ABC' + ABC$ was implemented directly using AND, OR, and NOT gates, producing four product terms combined by a 4-input OR gate. The simplified expression $AB + AC + BC$ was then implemented separately using only three 2-input AND gates and a 3-input OR gate, giving a circuit with fewer gates.

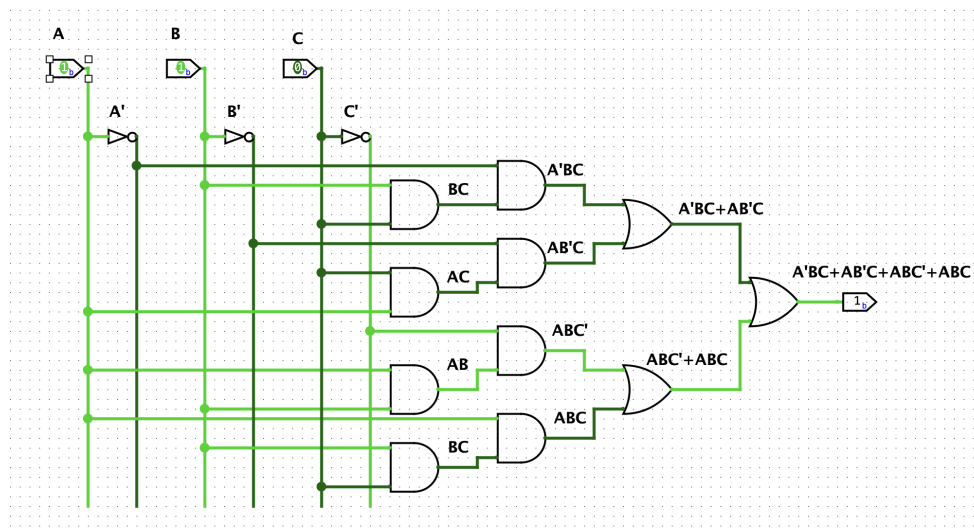


Figure 1: Circuit before simplification: $F = A'BC + AB'C + ABC' + ABC$

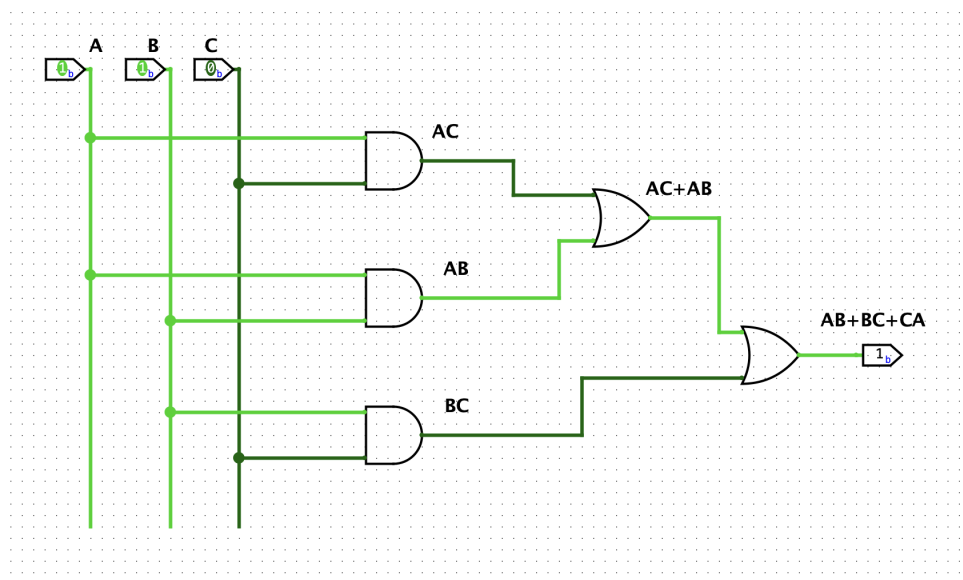


Figure 2: Circuit after simplification: $F = AB + AC + BC$

Result and Discussion

Both the unsimplified and simplified circuits were tested for all eight combinations of A, B, and C. The outputs obtained from both circuits matched each other and matched the expected values shown in the truth table above, confirming that the simplification is correct.

Conclusion

This experiment demonstrated that the Boolean expression $A'BC + AB'C + ABC' + ABC$ can be simplified to $AB + AC + BC$, reducing the number of gates required while preserving identical logical behavior.

Task 2: Simplification and Implementation of $F(A, B, C) = A(A + B)(A + B + C)$

Objectives

- To simplify the given Boolean expression using the absorption law.
- To implement the unsimplified and simplified expressions as logic circuits.
- To verify that both circuits produce identical outputs for all input combinations.

Apparatus

Digital logic circuit simulation software, AND/OR gate blocks, input switches, output indicator (LED).

Given Expression

$$F(A, B, C) = A(A + B)(A + B + C)$$

Simplification

$$\begin{aligned} F(A, B, C) &= A(A + B)(A + B + C) \\ &= A + AB + AC + AB + AB + ABC \\ &= A + AB + AC + ABC \\ &= A + AB + AC \quad (\text{since } ABC \text{ is absorbed by } AB) \\ &= A(1 + B + C) \\ &= A \end{aligned}$$

Truth Table Verification

A	B	C	$F = A$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Circuit Design

The unsimplified expression $A(A + B)(A + B + C)$ was implemented directly using two OR gates to form $(A + B)$ and $(A + B + C)$, followed by a 3-input AND gate combining

these with A . The simplified expression $F = A$ was then implemented as a direct wire connection from input A to the output.

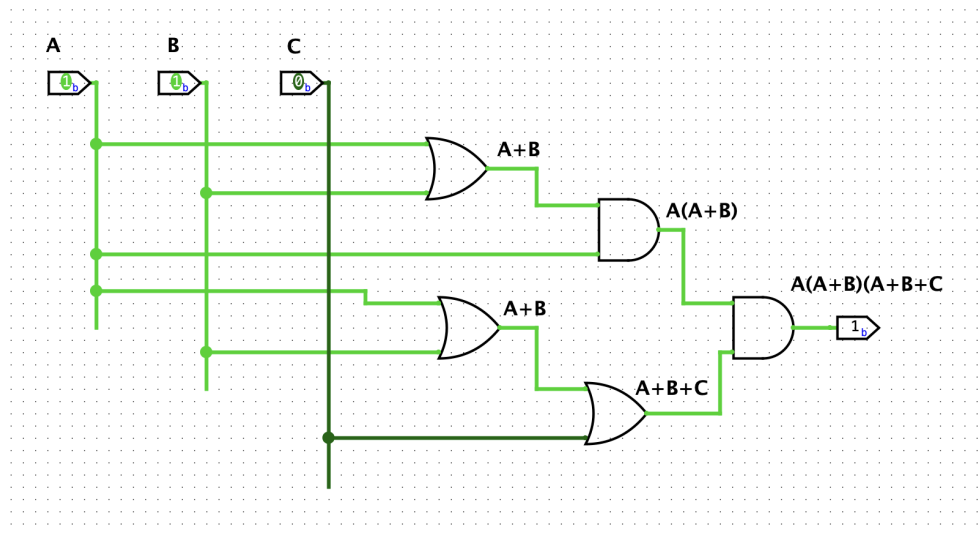


Figure 3: Circuit before simplification: $F = A(A + B)(A + B + C)$

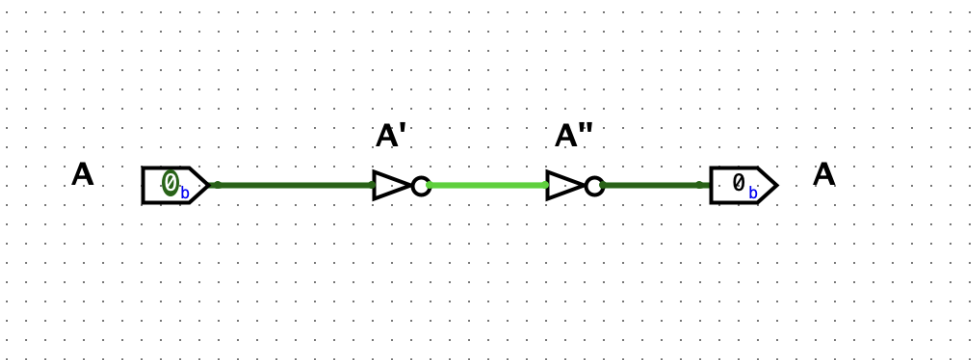


Figure 4: Circuit after simplification: $F = A$

Result and Discussion

Both circuits were tested for all eight combinations of A , B , and C . In every case, the output matched the value of A directly, confirming that the entire expression reduces to A regardless of the values of B and C .

Conclusion

This experiment demonstrated that repeated absorption reduces $A(A + B)(A + B + C)$ to the single variable A , eliminating the dependency on B and C entirely and greatly simplifying the circuit.

Task 3: Simplification and Implementation of $F(A, B, C) = (A + (BC)')'(AB + ABC)$

Objectives

- To simplify the given Boolean expression using De Morgan's theorem and complementation.
- To implement the unsimplified and simplified expressions as logic circuits.
- To verify that both circuits produce identical (constant) outputs for all input combinations.

Apparatus

Digital logic circuit simulation software, AND/OR/NOT gate blocks, input switches, output indicator (LED).

Given Expression

$$F(A, B, C) = (A + (BC)')'(AB + ABC)$$

Simplification

$$\begin{aligned} F(A, B, C) &= (A + (BC)')'(AB + ABC) \\ &= (A + (BC)')' \cdot AB \quad (\text{since } AB + ABC = AB(1 + C) = AB) \\ &= AB(A + (BC)')' \\ &= AB(A + B' + C')' \quad (\text{since } (BC)' = B' + C') \\ &= AB \cdot A'BC \quad (\text{De Morgan: } (A + B' + C')' = A'BC) \\ &= A \cdot B \cdot A' \cdot B \cdot C \\ &= (AA')(BB)C \\ &= 0 \quad (\text{since } AA' = 0). \end{aligned}$$

Truth Table Verification

A	B	C	$F = 0$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Circuit Design

The unsimplified expression was implemented directly, generating $(BC)'$ with a NAND gate, combining it with A through an OR gate followed by a NOT gate, generating AB and ABC separately and OR-ing them, and finally AND-ing the two halves together. The simplified expression $F = 0$ was implemented as the output tied permanently to logic LOW (ground).

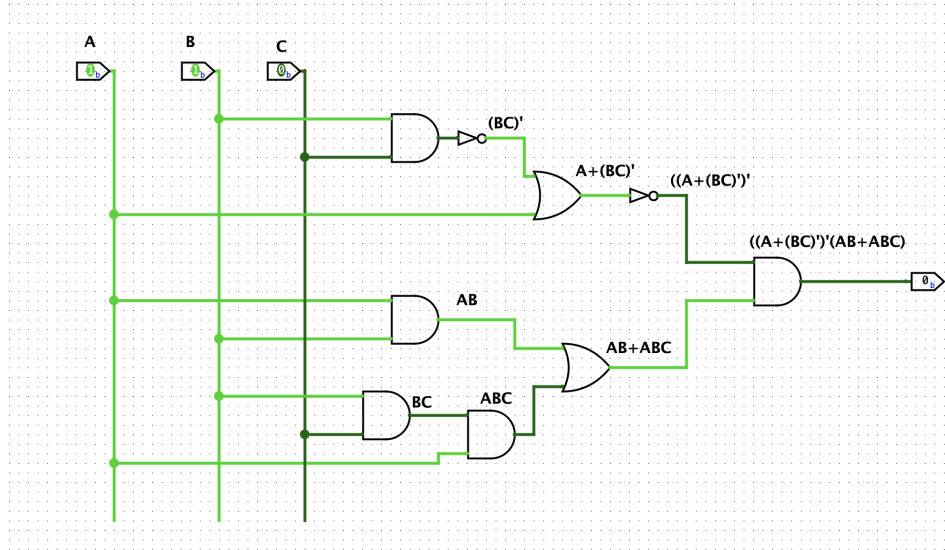


Figure 5: Circuit before simplification: $F = (A + (BC)')'(AB + ABC)$

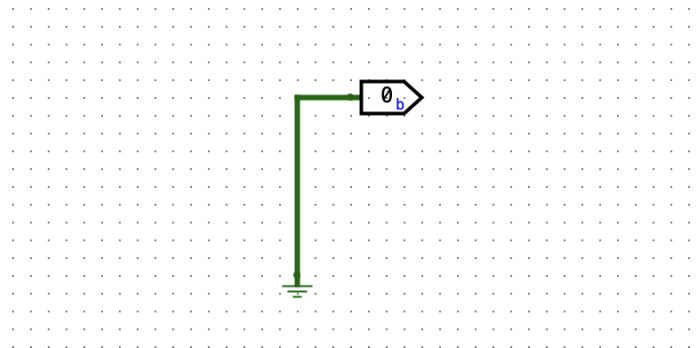


Figure 6: Circuit after simplification: $F = 0$

Result and Discussion

Both circuits were tested for all eight combinations of A, B, and C. The output remained LOW in every case for both the unsimplified and simplified circuits, confirming that the expression is a contradiction that always evaluates to 0.

Conclusion

This experiment demonstrated that the expression $(A + (BC)')'(AB + ABC)$ simplifies to the constant 0, since the two ANDed terms AB and $A'BC$ contain complementary literals (A and A') and can never both be true, illustrating how algebraic simplification can reveal a circuit that is logically always off.

Task 4: Simplification and Implementation of $F(A, B, C) = (B'(A + B) + (A + B)(A + B'))B'$

Objectives

- To simplify the given Boolean expression using distribution and absorption.
- To implement the unsimplified and simplified expressions as logic circuits.
- To verify that both circuits produce identical outputs for all input combinations.
- To understand the practical application of Boolean algebra in the design and optimization of digital logic circuits.

Apparatus

Digital logic circuit simulation software, AND/OR/NOT gate blocks, input switches, output indicator (LED).

Given Expression

$$F(A, B, C) = (B'(A + B) + (A + B)(A + B'))B'$$

Simplification

$$\begin{aligned} F(A, B, C) &= (B'(A + B) + (A + B)(A + B'))B' \\ &= (AB' + BB' + AA + AB' + AB + BB')B' \\ &= (AB' + AB' + AB)B' \quad (\text{since } BB' = 0, AA = A) \\ &= (A + AB + AB')B' \\ &= A(1 + B + B')B' \\ &= AB' \end{aligned}$$

Truth Table Verification

A	B	C	$F = AB'$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

Note: F does not depend on C in either the unsimplified or simplified form.

Circuit Design

The unsimplified expression was implemented directly using AND, OR, and NOT gates to build each of the bracketed terms $B'(A + B)$ and $(A + B)(A + B')$, OR-ing them together, and finally AND-ing the result with B' . The simplified expression $F = AB'$ was implemented using a single NOT gate (for B') and a single AND gate combining it with A .

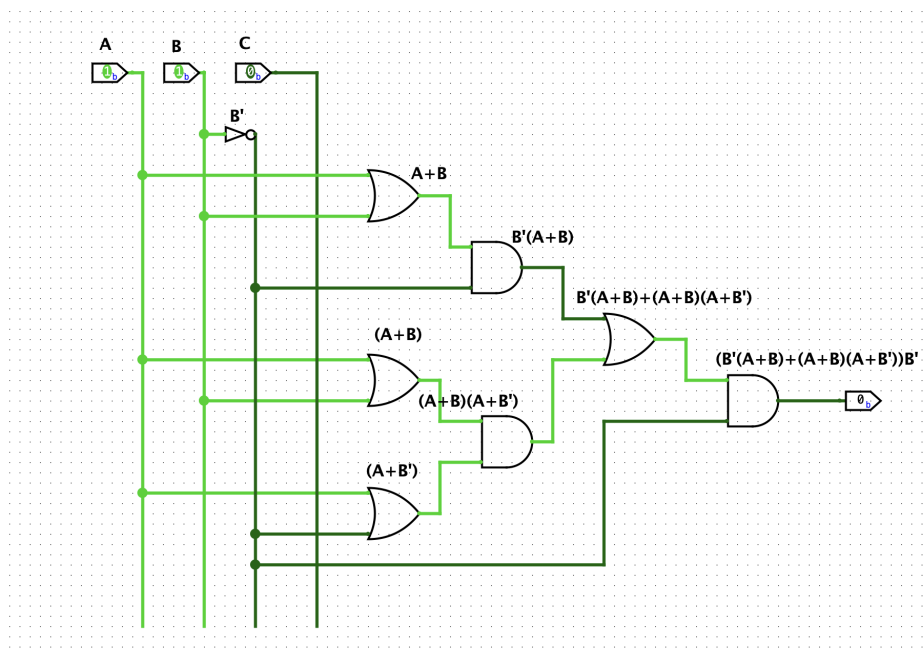


Figure 7: Circuit before simplification: $F = (B'(A + B) + (A + B)(A + B'))B'$

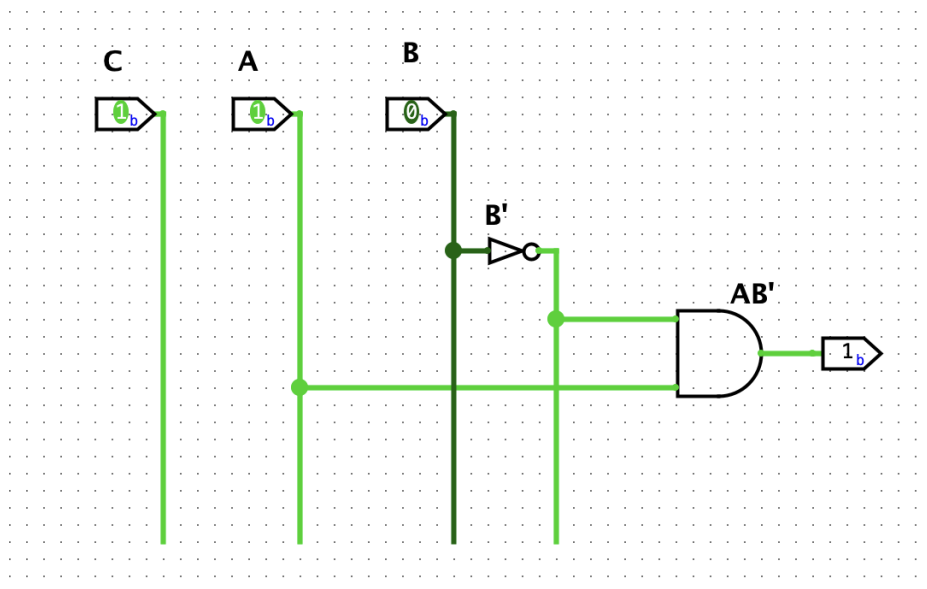


Figure 8: Circuit after simplification: $F = AB'$

Result and Discussion

Both circuits were tested for all eight combinations of A, B, and C. The outputs matched each other and matched the expected values in the truth table above, confirming that the simplified two-gate circuit is functionally identical to the original four-term circuit.

Conclusion

This experiment demonstrated that the expression $(B'(A + B) + (A + B)(A + B'))B'$ simplifies to AB' , reducing a circuit with two OR terms and multiple AND operations down to a single AND gate and a single inverter.

References

1. M. M. Mano, *Digital Design*, 5th ed. Upper Saddle River, NJ: Pearson, 2013.
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4. Logisim-evolution, Digital Logic Design Simulator, Documentation. Available: <https://github.com/logisim-evolution/logisim-evolution>